School of Engineering and Applied Science (SEAS), Ahmedabad University

ECE501: Digital Image Processing

Group Name: Humans.exe

Project 2: Person Retrieval

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I. Introduction

- **Goal:** Person Retrieval - finding all images of the same person in a given photo database.
- **Common Method:** Normally done with Machine Learning / Deep Learning.
- **Our Approach:** Use only Classical Digital Image Processing (DIP) Techniques.
- **Our Database:** 20 photos → 4 persons x 5 photos each.

II. Objectives

This week aims at improving the reliability and accuracy of the system by detecting and reducing the current defects. The main areas of concern are the enhancement of the accuracy of feature extraction and matching in order to guarantee more consistent person retrieval. The adjustment will be focused on the fine-tuning of the parameters, the optimization of the similarity measures, and the testing of the system in different conditions to measure the performance. It is hoped that through attentively studying the patterns of errors and optimization of the existing pipeline, it will be possible to minimize the number of mismatches, enhance stability, and obtain more resilient results. All in all, this week attempts to make the system nearer to the targeted accuracy and operational efficiency.

III. What has been done so far (Progress)

This has far been achieved to a great extent in regard to enhancing the performance and reliability of the system. The first attempt was aimed at determination of the main failures and bottlenecks in the current pipeline, especially in the extraction and matching of features. The analysis involved the presence of relevant problems like decreased accuracy, unreliable recovery, and different recovery under different conditions.

The feature extraction module was improved to produce more specific representations which would produce more accurate matching. Further parameter optimization was done to adjust thresholds, similarity computation, and accuracy of person retrieval task. Even the stability testing has been done beforehand to make sure that the improvements will be kept universal in regards to different scenarios. In general, such activities have reinforced the main workflow and preconditioned additional fine-tuning and growth.

IV. What is planned for next week

- Continue with possible corrections to the remaining problems in the system and reduce error detected during testing.
- Increase the quality of feature extraction and matching with parameter fine-tuning and enhancement of threshold selection.
- Carry out additional scenario testing to prove better and increase stability of the system.
- Test other refinement options, such as model adaptations, to minimize mismatches and enhance maximum accuracy.

References

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